

Physician-associated variation in chest X-ray prediction

RESEARCH QUESTION

Does including one attending physician's cases in training change a model's atelectasis predictions for that physician's held-out patients?

01

Motivation

02

Data & labels

03

Method

04

Results & next steps



Prediction target: atelectasis as stated in the radiology report

Labels come from reports, and reports come from physicians

So a model trained on report-derived labels could also absorb physician-associated patterns.



Labels are read from report text

CheXbert turns each radiology report into positive, negative, uncertain or not mentioned [2].



Every report is linked to an attending physician

Case mix and wording (e.g. “likely atelectasis”) can differ from one physician to another.



CNNs can pick up source-specific signals

Pneumonia CNNs generalised unevenly across hospitals and could even identify the source hospital [3].

RESEARCH QUESTION

If training includes studies linked to physician A, does the model perform differently on A’s held-out patients?

Why it matters

Scores on familiar sources may overstate how a model does on new ones.

Scope

A robustness check, not a judgement of physician quality.

Prior work compares sites; we compare training exposure

Holding the test patients fixed removes test-set differences from the comparison.



Generalisation across sites

Zech et al., PLOS Medicine 2018 [3]

Pneumonia CNNs performed differently on other hospitals' data and could recognise which hospital an image came from.



Labels from report text

Smit et al., EMNLP 2020 [2]

CheXbert labels 14 findings from report text as positive, negative, uncertain or blank (not mentioned).



Images, reports, providers

MIMIC-CXR v2.1.0; Johnson et al., 2019 [1]

De-identified radiographs with free-text reports; each study links to anonymised attending IDs.



Our angle: a paired, physician-level contrast

Same recipe, validation set and 109 test studies. Training sets are matched on size, positives and views; one includes 46 of A's studies, the other none.

Seen · with A

Unseen · no A

Same test

One course subset, 35 attending IDs, very uneven volume

Each study links one patient, its images, one report and an anonymised attending ID.

1,000

patients

3,351

studies

5,534

images

INSIDE ONE STUDY



Frontal image → model input

One per study: PA preferred, otherwise AP



Report → atelectasis label

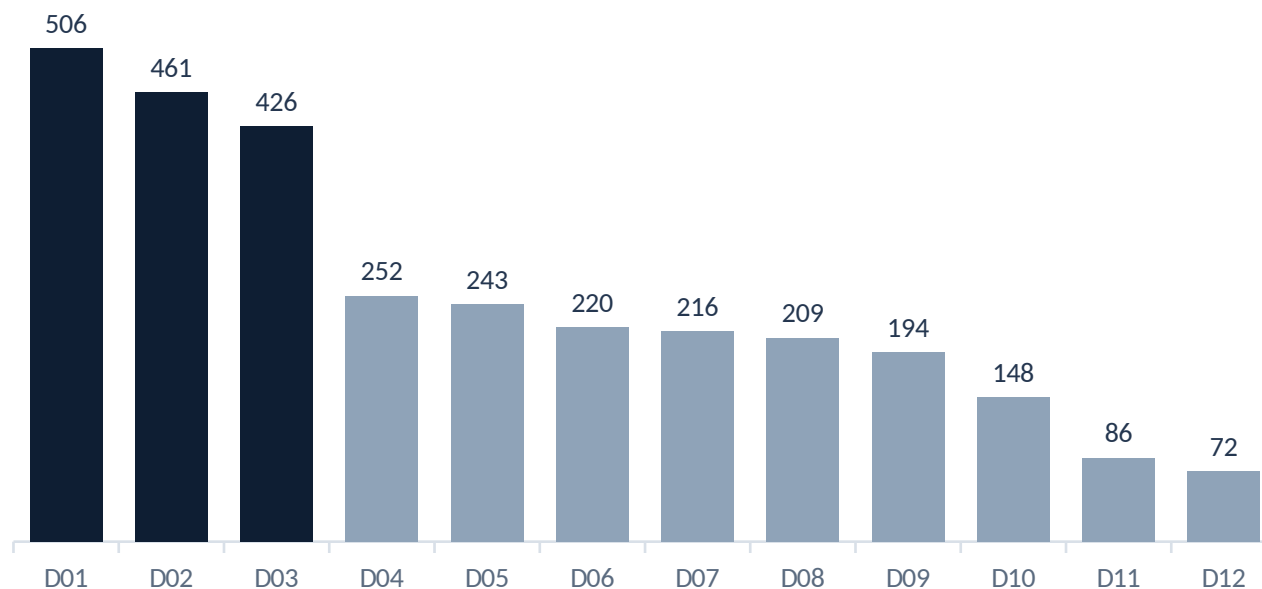
Used to make labels; never fed to the model



Attending ID → grouping only

Defines physician groups; never a model input

Studies per attending ID • top 12 of 35 (ranked)



42%

of studies: top 3 IDs

10

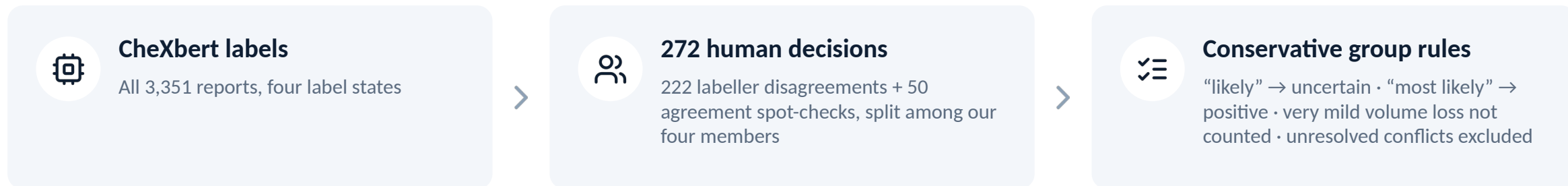
IDs with ≥ 100 patients

23

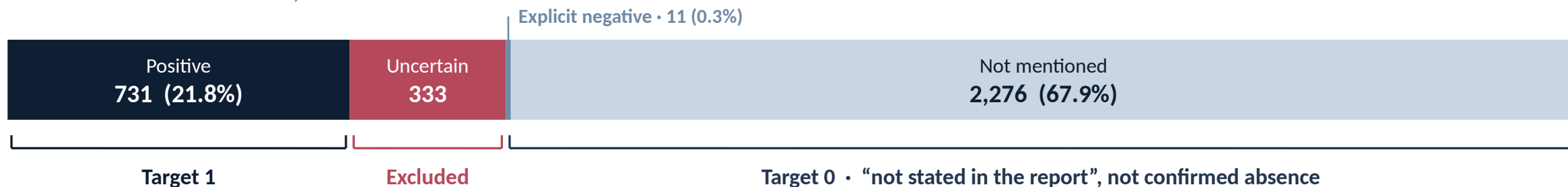
IDs share the last 318 studies

The target is atelectasis as stated in the report

Automatic labels, human decisions where labellers disagreed, then fixed group rules.



FINAL REPORT LABEL · ALL 3,351 STUDIES



Eligible after filtering

uncertain or ambiguous removed · frontal view needed

2,904 studies

940 patients

709 positive (24%)

1,728 / 1,176 AP / PA views

Matched training sets, with and without physician A's cases

Split by patient, match the training sets, share validation, and test both models on the same patients.

- 1 Split patients 60/20/20, no overlap
- 2 Build two matched training sets
- 3 Train ResNet-18 $\times 2$, same recipe
- 4 Pick epochs on shared validation
- 5 Test both on A's held-out patients

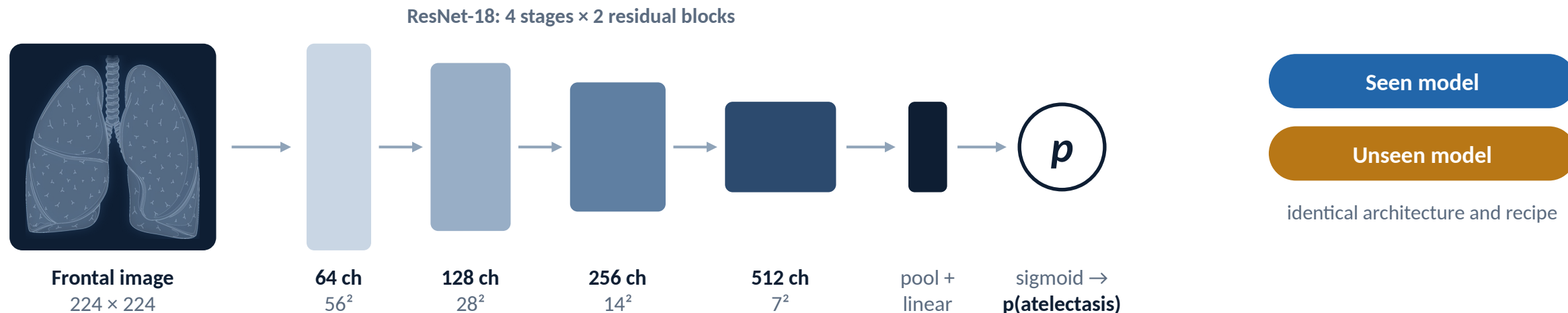
SET	STUDIES	PATIENTS	POSITIVE	PA / AP
Seen training includes physician A	46 256	167	59	110 / 146
Unseen training excludes physician A (0 studies)	256	171	59	110 / 146
Shared validation no A cases · selects epochs	128	85	26	51 / 77
Common test cohort all from A's held-out patients	109	55	39	41 / 68

Matched
size, positives and
PA/AP views

A = attending ID with the most eligible patients, fixed before training. The 256 / 128 budget was set before any training or test evaluation; full-size sets (1,409 per group) are kept for the final. Training sets may share non-A studies; no patient crosses train / validation / test.

An image-only ResNet-18, selected on validation only

Both models share the backbone and the training recipe; only their training sets differ.



Preprocessing

- DICOM rescale; invert MONOCHROME1
- Clip to 0.5–99.5th percentile
- Keep aspect ratio, pad to 224 $\times$ 224
- ImageNet normalisation; no augmentation



Training

- ImageNet init; all layers fine-tuned
- AdamW, lr 1e-4, weight decay 1e-4
- Batch 32, mixed precision, 5 epochs
- Keep epoch with lowest validation BCE



Evaluation

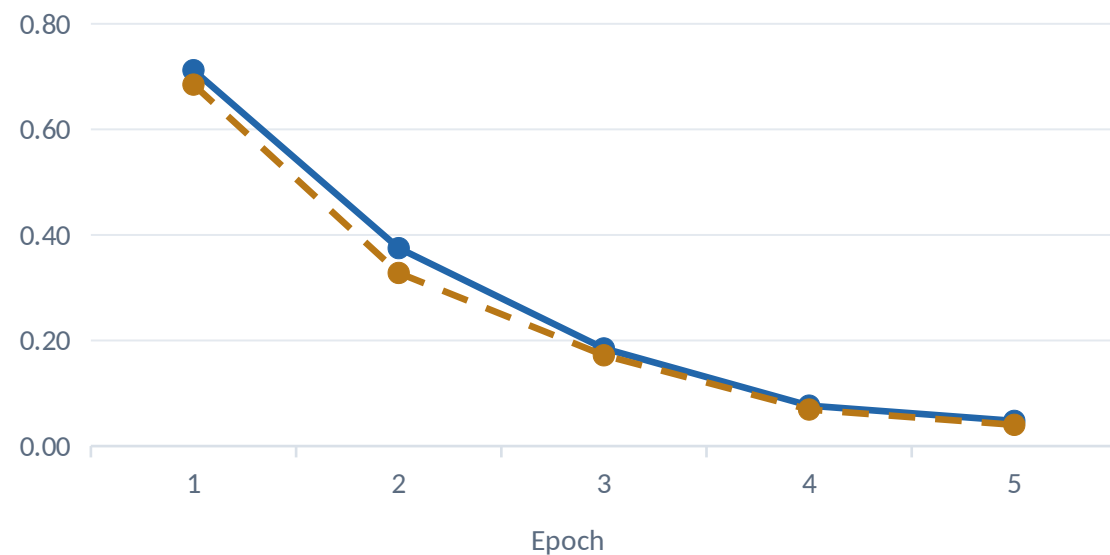
- AUROC (primary), AUPRC, Brier score
- Δ = seen – unseen on the same studies
- 95% CI: 1,000 patient-paired bootstraps
- Test set never used for any choice

Validation loss chose epoch 3 (seen) and epoch 4 (unseen)

Training loss kept falling; validation loss rose again after its minimum, so later epochs were not kept.

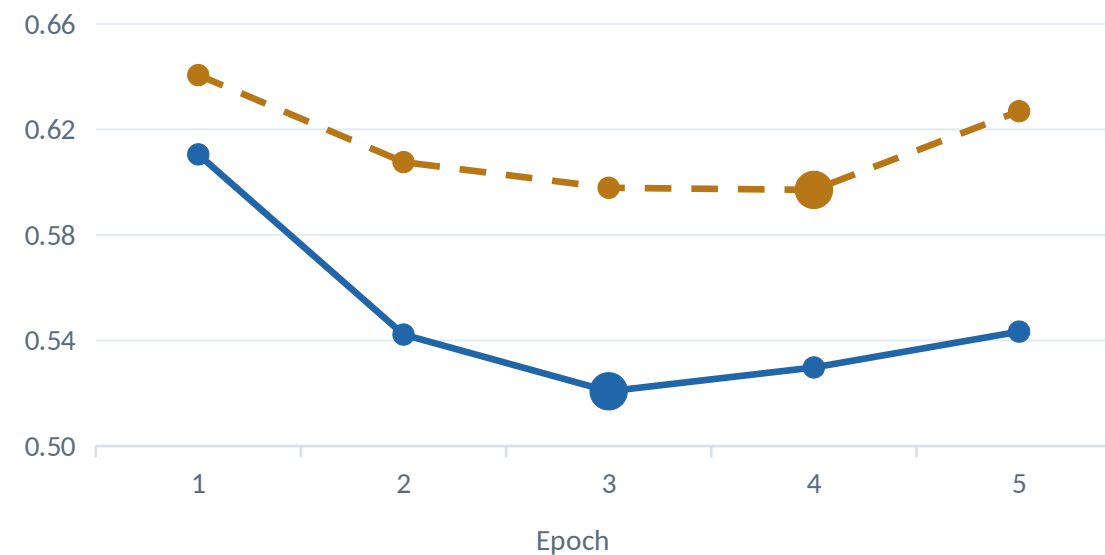
—●— Seen (includes A) -●- Unseen (excludes A)

Training BCE



Seen 0.712 → 0.048 **Unseen** 0.685 → 0.040

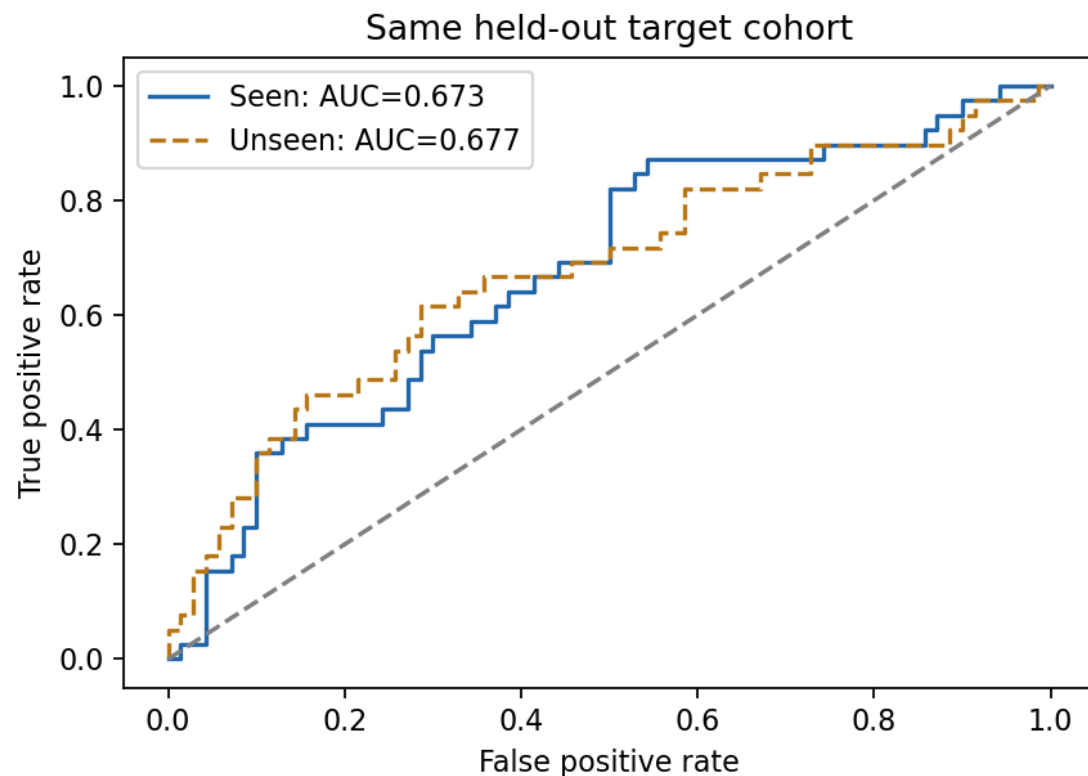
Shared validation BCE (no A cases)



Kept: seen epoch 3 • 0.521 unseen epoch 4 • 0.597

Same 109 test studies: AUROC 0.673 (seen) vs 0.677 (unseen)

Both models beat a constant-probability baseline, and their ROC curves cross repeatedly.



ROC on the common test cohort (109 studies · 55 patients · 39 positive). Figure: results/midterm_v1/results.png

Metric	Seen	Unseen	Constant baseline
AUROC ↑	0.673	0.677	0.500
AUPRC ↑	0.511	0.573	0.358
Brier ↓	0.222	0.219	0.246

HOW TO READ THE METRICS

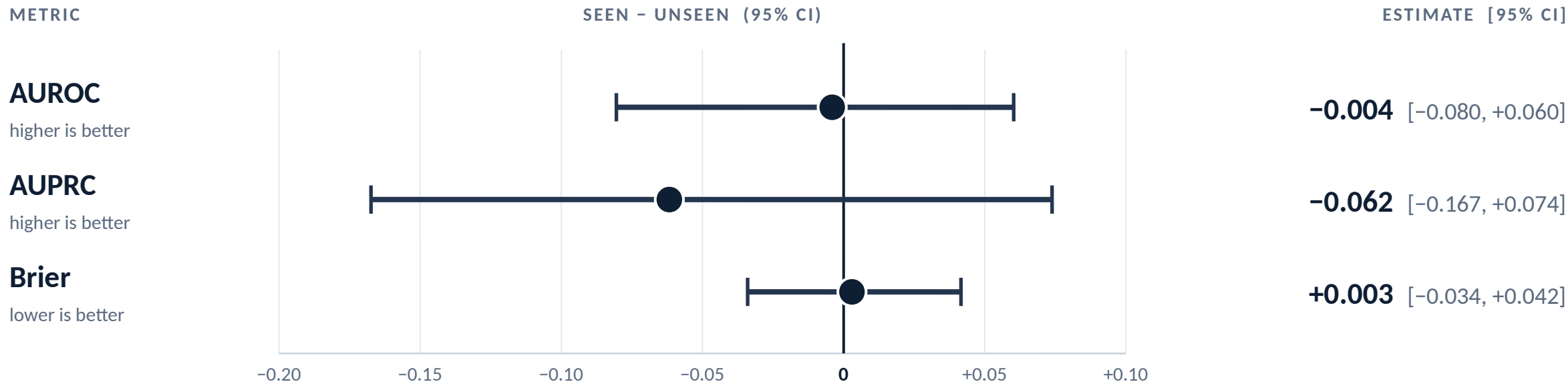
AUROC chance that a random positive study is ranked above a random negative one (0.5 = chance). Not accuracy.

AUPRC precision-recall summary; its chance level is the prevalence, $39/109 = 0.358$.

Brier mean squared error of the predicted probabilities; lower is better.

All three paired 95% intervals include zero

Seen – unseen on the same test patients: no stable difference in this pilot.



No stable difference

Training with A's 46 studies gave no detectable difference on A's patients in this pilot.



Not proof of equivalence

AUROC differences from -0.080 to +0.060 remain compatible with these data.



Uncertainty is partial

Intervals resample test patients only, not retraining with another seed.

What the pilot shows, and what comes next

A working paired pipeline on real images, with uncertainty still too wide for a firm conclusion.



Limits of this pilot

- One physician group, one training seed
- Small samples: 256 training studies per group, 55 test patients
- Report-derived labels, not an image gold standard
- Case mix not fully matched (e.g. 167 vs 171 training patients)



Plan for the final project

- 1 Repeat across more attending IDs and several seeds
- 2 Train on the prepared full-size sets (1,409 studies per group)
- 3 Audit a random sample of “not mentioned” labels
- 4 Examine case mix (views, patient overlap) and 512 px inputs



Fixed in advance: test cohorts, selection rules and metrics — no re-picking physicians or test sets to improve the numbers.

References and credits

Backup slide — not part of the 7-minute talk.

- [1] Johnson AEW, Pollard TJ, Berkowitz SJ, et al. MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports. Scientific Data 6, 317 (2019). Data: MIMIC-CXR v2.1.0, PhysioNet.
physionet.org/content/mimic-cxr/2.1.0

- [2] Smit A, Jain S, Rajpurkar P, et al. CheXbert: combining automatic labelers and expert annotations for accurate radiology report labeling using BERT. EMNLP 2020.
arxiv.org/abs/2004.09167

- [3] Zech JR, Badgeley MA, Liu M, et al. Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study. PLOS Medicine 15(11): e1002683 (2018).
doi.org/10.1371/journal.pmed.1002683

- [4] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. CVPR 2016.
arxiv.org/abs/1512.03385

Project results: experiment run on 27 Sep 2026 (one physician group, one seed); metrics, figure and configuration in results/midterm_v1.
github.com/Grayve123/BN5212_group

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